

5.35 Solution: (a) From Problem 5.13, we know that if the current density is in the form of

$$\mathbf{K} = \mathbf{j} \times \mathbf{r} \delta(r - a),$$

where $\mathbf{j}$ is in the z -direction, then the magnetic field in the sphere is

$$\mathbf{B} = \frac{2}{3} \mu a \mathbf{j}.$$

Since the magnetic induction is a constant of B_0 , we must have

$$|\mathbf{j}| = \frac{3B_0}{2\mu a},$$

and

$$\mathbf{K} = \frac{3B_0}{2\mu a} \sin \theta \hat{\phi}.$$

The vector potential is then given by

$$A_\phi = \frac{\mu |\mathbf{j}| a^3}{3} \frac{r_{<}}{r_{>}^2} \sin \theta = \frac{B_0 a^2}{2} \frac{r_{<}}{r_{>}^2} \sin \theta.$$

(b) With the steady current density, we can see that $A_\phi e^{i\phi}$ is proportional to $Y_{11}(\theta, \phi)$. Then, we can expand it in spherical Bessel functions,

$$-\sqrt{\frac{3}{8\pi}} e^{i\phi} A_\phi(\mathbf{x}, t = 0) = \int_0^{+\infty} \tilde{A}(k) j_1(kr) dk \cdot Y_{11}(\theta, \phi),$$

where

$$\begin{aligned} \tilde{A}(k) &= \frac{2k^2}{\pi} \int_0^{+\infty} \frac{B_0 a^2}{2} \frac{r_{<}}{r_{>}^2} j_1(kr) r^2 dr \\ &= \frac{B_0 k^2 a^2}{\pi} \left[\frac{1}{a^2} \int_0^a r^3 j_1(kr) dr + a \int_a^{+\infty} j_1(kr) dr \right] \\ &= \frac{B_0 k^2 a^2}{\pi} \left[\frac{1}{k^4 a^2} \int_0^{ka} x^3 j_1(x) dx + \frac{a}{k} \int_{ka}^{+\infty} j_1(x) dx \right] \\ &= \frac{B_0 k^2 a^2}{\pi} \left[-\frac{1}{k^4 a^2} ((k^2 a^2 - 3) \sin(ka) + 3ka \cos(ka)) + \frac{a}{k} \frac{\sin(ka)}{ka} \right] \\ &= \frac{3B_0 a^2}{\pi} \left(\frac{\sin(ka)}{k^2 a^2} - \frac{\cos(ka)}{ka} \right) \\ &= \frac{3B_0 a^2}{\pi} j_1(ka). \end{aligned}$$

For $t > 0$, if we express the vector potential as

$$-\sqrt{\frac{3}{8\pi}} e^{i\phi} A_\phi(\mathbf{x}, t) = \int_0^{+\infty} \tilde{A}(k, t) j_1(kr) dk \cdot Y_{11}(\theta, \phi),$$

then $\tilde{A}(k, t)$ satisfies the following equation,

$$\mu \sigma \frac{\partial \tilde{A}(k, t)}{\partial t} = -k^2 \tilde{A}(k, t),$$

with $\tilde{A}(k, t = 0) = \tilde{A}(k)$. Here, we have used the fact that $j_1(kr)Y_{11}(\theta, \phi)$ satisfies the Helmholtz equation

$$(\nabla^2 + k^2)\psi(\mathbf{x}) = 0,$$

in spherical coordinate system. It is easy to see that

$$\tilde{A}(k, t) = \tilde{A}(k)e^{-k^2t/\mu\sigma}.$$

Then,

$$\begin{aligned} A_\phi(\mathbf{x}, t) &= \frac{3B_0a^2}{\pi} \sin \theta \int_0^{+\infty} e^{-k^2t/\mu\sigma} j_1(ka) j_1(kr) dk \\ &= \frac{3B_0a}{\pi} \sin \theta \int_0^{+\infty} e^{-k^2t/\mu\sigma a^2} j_1(k) j_1\left(\frac{kr}{a}\right) dk. \end{aligned}$$

Let $\nu = (\mu\sigma a^2)^{-1}$, we will finally have

$$A_\phi(\mathbf{x}, t) = \frac{3B_0a}{\pi} \sin \theta \int_0^{+\infty} e^{-\nu tk^2} j_1(k) j_1\left(\frac{kr}{a}\right) dk.$$

Near the center of the sphere, $r \ll a$, and we can approximate $j_1(x) \sim x/3$. The vector potential can be evaluated as

$$A_\phi(\mathbf{x}, t) = \frac{3B_0a}{\pi} \sin \theta \cdot \frac{r}{3a} \int_0^{+\infty} e^{-\nu tk^2} k j_1(k) dk = \frac{B_0}{\pi} r \sin \theta \int_0^{+\infty} e^{-\nu tk^2} k j_1(k) dk.$$

Since

$$j_1(k) = \frac{\sin k}{k^2} - \frac{\cos k}{k},$$

using Gradshteyn and Ryzhik, 8th ed., 3.952.6, pg. 507, we know

$$\int_0^{+\infty} e^{-\nu tk^2} \frac{\sin k}{k} dk = \frac{\pi}{2} \Phi\left(\frac{1}{2\sqrt{\nu t}}\right).$$

The integration with $\cos k$ is elementary,

$$\int_0^{+\infty} e^{-\nu tk^2} \cos k dk = \frac{1}{2} \sqrt{\frac{\pi}{\nu t}} \exp\left(-\frac{1}{4\nu t}\right),$$

we then have the vector potential as

$$\begin{aligned} A_\phi(\mathbf{x}, t) &= \frac{B_0}{\pi} \cdot \left[\frac{\pi}{2} \Phi\left(\frac{1}{\sqrt{4\nu t}}\right) - \frac{1}{2} \sqrt{\frac{\pi}{\nu t}} \exp\left(-\frac{1}{4\nu t}\right) \right] \cdot r \sin \theta \\ &= B_0 \left[\Phi\left(\frac{1}{\sqrt{4\nu t}}\right) - \frac{1}{\sqrt{\pi\nu t}} \exp\left(-\frac{1}{4\nu t}\right) \right] \frac{r \sin \theta}{2}. \end{aligned}$$

In spherical coordinates, the magnetic induction is given by

$$\begin{aligned} \mathbf{B}(0, t) &= \nabla \times \mathbf{A} \\ &= \hat{r} \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\phi) - \hat{\theta} \frac{1}{r} \frac{\partial}{\partial r} (r A_\phi) \end{aligned}$$

$$\begin{aligned}
&= B_0 \left[\Phi \left(\frac{1}{\sqrt{4\nu t}} \right) - \frac{1}{\sqrt{\pi\nu t}} \exp \left(-\frac{1}{4\nu t} \right) \right] (\hat{r} \cos \theta - \hat{\theta} \sin \theta) \\
&= B_0 \left[\Phi \left(\frac{1}{\sqrt{4\nu t}} \right) - \frac{1}{\sqrt{\pi\nu t}} \exp \left(-\frac{1}{4\nu t} \right) \right] \hat{z},
\end{aligned}$$

where we have only the z -component.

(c) The magnetic energy is

$$W_m = \frac{1}{2} \int \mathbf{B} \cdot \mathbf{H} d^3x = \frac{1}{2\mu} \int (\nabla \times \mathbf{A}) \cdot (\nabla \times \mathbf{A}) d^3x.$$

From the vector relation

$$\nabla \cdot (\mathbf{a} \times \mathbf{b}) = \mathbf{b} \cdot (\nabla \times \mathbf{a}) - \mathbf{a} \cdot (\nabla \times \mathbf{b}),$$

and let $\mathbf{a} = \mathbf{A}$ and $\mathbf{b} = \nabla \times \mathbf{A}$, we have

$$(\nabla \times \mathbf{A}) \cdot (\nabla \times \mathbf{A}) = \nabla \cdot (\mathbf{A} \times (\nabla \times \mathbf{A})) + \mathbf{A} \cdot (\nabla \times (\nabla \times \mathbf{A})) = \nabla \cdot (\mathbf{A} \times (\nabla \times \mathbf{A})) + \mathbf{A} \cdot \nabla (\nabla \cdot \mathbf{A}) - \mathbf{A} \cdot \nabla^2 \mathbf{A}.$$

Upon integration, the first term becomes a surface integral and is zero, therefore can be dropped. Also, $\nabla \cdot \mathbf{A} = 0$, as $\mathbf{A}$ has only ϕ -component which does not depend on ϕ . Finally, $\nabla^2 \mathbf{A} = \mu\sigma \frac{\partial \mathbf{A}}{\partial t}$. The magnetic energy now becomes

$$W_m = -\frac{1}{2\mu} \int \mathbf{A} \cdot \mu\sigma \frac{\partial \mathbf{A}}{\partial t} d^3x.$$

Using result of part (b), we can write the magnetic energy as

$$\begin{aligned}
W_m &= -\frac{1}{2\mu} \int d^3x \frac{3B_0 a}{\pi} \sin \theta \int_0^{+\infty} e^{-\nu t k_1^2} j_1(k_1) j_1 \left(\frac{k_1 r}{a} \right) dk_1 \\
&\quad \cdot \mu\sigma \frac{\partial}{\partial t} \left(\frac{3B_0 a}{\pi} \sin \theta \int_0^{+\infty} e^{-\nu t k_2^2} j_1(k_2) j_1 \left(\frac{k_2 r}{a} \right) dk_2 \right) \\
&= \frac{1}{2\mu} \int dk_1 \int dk_2 \frac{9B_0^2 a^2}{\pi^2} \mu\sigma \nu k_2^2 j_1(k_1) j_1(k_2) e^{-\nu t(k_1^2 + k_2^2)} \\
&\quad \cdot \int d^3x \sin^2 \theta j_1 \left(\frac{k_1 r}{a} \right) j_1 \left(\frac{k_2 r}{a} \right).
\end{aligned}$$

The integral in space can be performed as

$$\begin{aligned}
\int d^3x \sin^2 \theta j_1 \left(\frac{k_1 r}{a} \right) j_1 \left(\frac{k_2 r}{a} \right) &= 2\pi \int_{-1}^1 \sin^2 \theta d(\cos \theta) \int_0^{+\infty} r^2 j_1 \left(\frac{k_1 r}{a} \right) j_1 \left(\frac{k_2 r}{a} \right) dr \\
&= 2\pi \cdot \frac{4}{3} \cdot \frac{\pi}{2(k_1/a)^2} \delta \left(\frac{k_1}{a} - \frac{k_2}{a} \right) = \frac{4\pi^2}{3} \frac{a^3}{k_1^2} \delta(k_1 - k_2).
\end{aligned}$$

Then, we can see that the magnetic energy is

$$\begin{aligned}
W_m &= \frac{1}{2\mu} \cdot \frac{9B_0^2 a^2}{\pi^2} \cdot \frac{4\pi^2}{3} \int dk_1 \int dk_2 \frac{k_2^2}{a^2} \cdot \frac{a^3}{k_1^2} \delta(k_1 - k_2) j_1(k_1) j_1(k_2) e^{-\nu t(k_1^2 + k_2^2)} \\
&= \frac{6B_0^2 a^3}{\mu} \int_0^{+\infty} e^{-2\nu t k^2} [j_1(k)]^2 dk.
\end{aligned} \tag{1}$$

By an integration variable change, $k = u/\sqrt{2\nu t}$, the integral can be expressed as

$$W_m = \frac{6B_0^2 a^3}{\mu} \frac{1}{\sqrt{2\nu t}} \int_0^{+\infty} e^{-u^2} \left[j_1 \left(\frac{u}{\sqrt{2\nu t}} \right) \right]^2 du.$$

For $x \ll 1$, $j_1(x) \sim x/3$. Therefore, in the limit of $\nu t \gg 1$, the integral becomes asymptotically

$$W_m \rightarrow \frac{6B_0^2 a^3}{\mu} \frac{1}{(2\nu t)^{3/2}} \int_0^{+\infty} \frac{u^2}{9} e^{-u^2} du = \frac{6B_0^2 a^3}{\mu} \frac{1}{2\sqrt{2}(\nu t)^{3/2}} \frac{\sqrt{\pi}}{36} = \frac{\sqrt{2\pi} B_0^2 a^3}{24\mu(\nu t)^{3/2}}.$$

We could also cut off the integral in Eq. (1) at $1/\sqrt{\nu t}$ and treat $e^{-2\nu t k^2}$ as 1 in this integration range. By this scheme, we will get the same asymptotic behavior, but with a different prefactor.

(d) Applying the same approximation as in part (c) to the vector potential, we will have

$$\begin{aligned} A_\phi &= \frac{3B_0 a}{\pi} \sin \theta \frac{1}{\sqrt{\nu t}} \int_0^{+\infty} e^{-u^2} j_1 \left(\frac{u}{\sqrt{\nu t}} \right) j_1 \left(\frac{u}{\sqrt{\nu t}} \frac{r}{a} \right) du \\ &\approx \frac{3B_0}{\pi} \sin \theta \frac{r}{(\nu t)^{3/2}} \int_0^{+\infty} \frac{u^2}{9} e^{-u^2} du \\ &= \frac{B_0}{12\sqrt{\pi}(\nu t)^{3/2}} r \sin \theta, \end{aligned}$$

which has the same asymptotic behavior as the magnetic energy. The magnetic induction is

$$\begin{aligned} \mathbf{B} &= \nabla \times \mathbf{A} \\ &= \hat{r} \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\phi) - \hat{\theta} \frac{1}{r} \frac{\partial}{\partial r} (r A_\phi) \\ &= \frac{B_0}{6\sqrt{\pi}(\nu t)^{3/2}} (\hat{r} \cos \theta - \hat{\theta} \sin \theta) = \frac{B_0}{6\sqrt{\pi}(\nu t)^{3/2}} \hat{z}, \end{aligned}$$

which is a constant at a specific time t when $r \ll R = a\sqrt{\nu t}$ and the approximation holds.

If $r \gg R$, the above approximation will fail. Instead, we make an integration variable change,

$$y = \frac{u}{\sqrt{\nu t}} \frac{r}{a}, \quad \text{or} \quad u = \frac{a\sqrt{\nu t}}{r} y,$$

and

$$A_\phi = \frac{3B_0 a}{\pi} \sin \theta \frac{1}{\sqrt{\nu t}} \int_0^{+\infty} \exp \left(-\frac{a^2 \nu t}{r^2} y^2 \right) j_1 \left(\frac{a}{r} y \right) j_1(y) \frac{a\sqrt{\nu t}}{r} dy.$$

Now, we can approximate $j_1(ay/r)$ as $ay/3r$, and use the result from part (b) to express the vector potential as

$$\begin{aligned} A_\phi &= \frac{B_0 a^3}{\pi r^2} \sin \theta \left[\frac{\pi}{2} \Phi \left(\frac{r}{2a\sqrt{\nu t}} \right) - \frac{1}{2} \sqrt{\frac{\pi r^2}{a^2 \nu t}} \exp \left(-\frac{r^2}{4a^2 \nu t} \right) \right] \\ &= \frac{B_0 a^3}{2r^2} \sin \theta \left[\Phi \left(\frac{r}{2a\sqrt{\nu t}} \right) - \frac{r}{a} \frac{1}{\sqrt{\pi \nu t}} \exp \left(-\frac{r^2}{4a^2 \nu t} \right) \right]. \end{aligned}$$

For $r \gg R$, we can see that the vector potential becomes

$$A_\phi = \frac{B_0 a^3}{2r^2} \sin \theta,$$

which is the result of part (a). This is reasonable, as when the current is cut off, the decay of the vector potential can only be propagated with a finite speed. For larger distance, the field will stay the same as the static result until a later time.